AI-POWERED JOB RECRUITMENT & CANDIDATE MANAGEMENT SYSTEM

1. PROJECT OVERVIEW

The AI-Powered Job Recruitment & Candidate Management System is a Salesforce-based application designed to streamline the recruitment process using Salesforce Admin tools and Agentforce AI.

The system helps recruiters and hiring managers manage candidate applications, automate approval workflows, and generate intelligent insights using AI.

Instead of manual candidate screening, the system uses AI-generated summaries, hiring risk analysis, and approval guidance to support hiring decisions.

This project was implemented entirely using Salesforce declarative tools, without using Apex development.

2. OBJECTIVES OF THE PROJECT

The main objective of the AI-Powered Job Recruitment & Candidate Management System is to streamline and automate the recruitment process using Salesforce and AI capabilities. The system is designed to improve efficiency, reduce manual effort, and support data-driven hiring decisions.

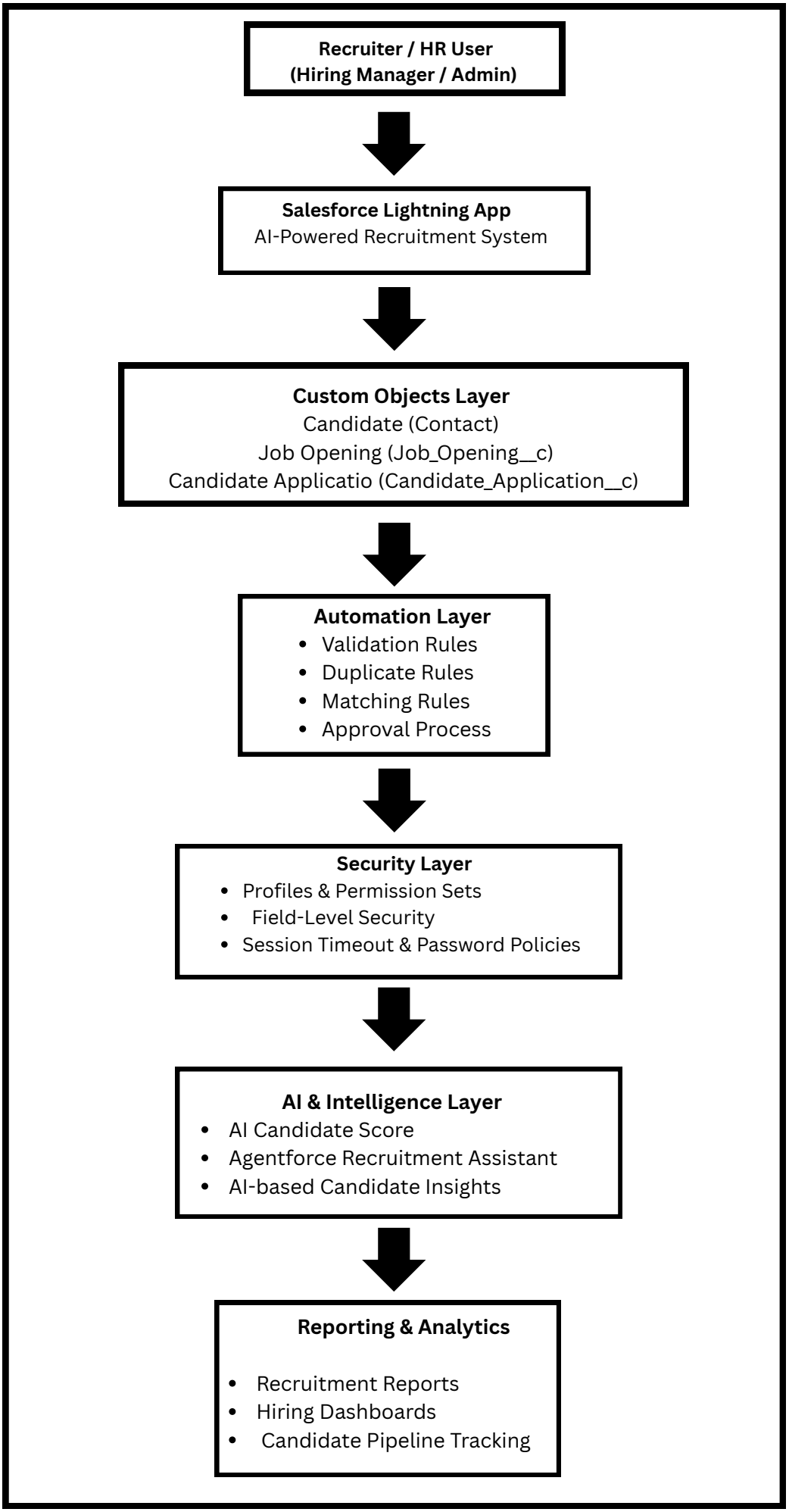
The key objectives of the project include:

• To develop a centralized platform for managing candidate information and job openings.
• To automate the candidate application process using Salesforce declarative tools.
• To implement an approval workflow for high salary candidate offers.
• To integrate Agentforce AI for generating intelligent candidate insights.
• To provide AI-generated candidate summaries for quick evaluation.
• To analyze potential hiring risks using AI-based analysis.
• To assist hiring managers in making better approval decisions using AI recommendations.
• To reduce manual screening and improve recruitment efficiency.

3.SCOPE OF THE PROJECT

• The system focuses on managing the recruitment process within a Salesforce environment.
• It allows recruiters to create and manage candidate profiles and job openings.
• The system supports candidate application tracking, linking candidates with specific job openings.
• It includes an automated approval process for candidate offers exceeding a defined salary threshold.
• The system integrates Agentforce AI to generate candidate summaries and hiring insights.
• AI is used to perform hiring risk analysis and provide recommendations for approval decisions.
• The platform enables recruiters and hiring managers to make data-driven recruitment decisions.
• The system demonstrates how Salesforce declarative tools and AI capabilities can be used to build an intelligent recruitment management system.
•The project scope is limited to Salesforce configuration, automation, and AI integration, without using custom Apex development.

4. SYSTEM ARCHITECTURE



5. SALESFORCE DEVELOPMENT – BACKEND CONFIGURATION

5.1 Custom Objects Created

The following custom objects were created:

OBJECTS	FIELDS:
Candidate Stores candidate information.	• Candidate Name • Email • Phone • Experience Level • Skills
Job Opening Stores job position details.	• Job Title • Department • Salary Range • Vacancies • Job Status
Candidate Application This object links candidates with job openings.	• Application Date • AI Candidate Score • Offer Amount • Application Status • Approval Status • Candidate (Lookup) • Job Opening (Lookup

6. FIELDS AND RELATIONSHIPS

Relationships implemented:
Candidate → Candidate Application **(Lookup)**
Job Opening → Candidate Application **(Master-Detail)**
This structure allows tracking which candidate applied for which job opening.

7. VALIDATION RULES

Validation rules were created to ensure data quality.
Activity 1: Offer Amount Validation (Candidate Application)
Objective: Ensure the offer amount is greater than zero.

Formula:
Offer_Amount__c <= 0
Error Message:
Offer amount must be greater than zero.
Result:
Prevents users from entering zero or negative offer amounts.

Activity 2: Job Vacancies Validation (Job Opening)

Objective:

Ensure the number of vacancies is not negative.

Formula:

Vacancies__c < 0

Error Message:

Vacancies cannot be negative.

Result:

Prevents invalid vacancy values from being saved.

Activity 3: AI Candidate Score Validation (Candidate Application)

Objective:

Ensure AI candidate score does not exceed 100.

Formula:

AI__Candidate_Score__c > 100

Error Message:

AI Candidate score cannot exceed 100.

Result:

Ensures the candidate score remains within the valid range.

8. APPROVAL PROCESS IMPLEMENTATION

An approval process was implemented to manage high salary candidate offers. If the offer amount exceeds a predefined threshold, the candidate application is automatically submitted for approval to the hiring manager.

Approval Criteria:

Offer Amount > **1,000,000**

The hiring manager reviews the application and decides whether to approve or reject the candidate offer.

Approval Actions:

• If approved → Approval Status is updated to Approved

• If rejected → Approval Status is updated to Rejected

This process ensures proper review and control over high salary hiring decisions.

9. EMAIL TEMPLATES AND ALERTS

Email templates and alerts were configured to automate communication during the recruitment process. These alerts notify recruiters and hiring managers about important candidate activities such as high-priority applications and approval requests.

Email Templates Created

1. High Salary Candidate Offer Approval Notification

This email template sends an alert to the hiring manager when the offer amount exceeds the defined threshold and requires approval.

Example content:

Candidate Name: {!Candidate_Application__c.Candidate__r.Name}

Offer Amount: {!Candidate_Application__c.Offer_Amount__c}

A candidate offer exceeds the approval limit and requires manager approval

2.High Priority Candidate Application Notification
This email template notifies the recruiter when a candidate with a high AI candidate score submits an application.

Example content:
Candidate Name: {!Candidate_Application__c.Candidate__r.Name}
Job Opening: {!Candidate_Application__c.Job_Opening__r.Name}
AI Candidate Score: {!Candidate_Application__c.AI_Candidate_Score__c}
A high priority candidate application has been detected.
Please review the candidate application immediately.

3. Candidate Application Follow-Up Reminder
This template reminds recruiters to follow up on candidate applications that remain pending for a long time.
Example content:
Candidate Name: {!Candidate_Application__c.Candidate__r.Name}
Application Status: Pending
Reminder: This candidate application has been pending for review.
Please take necessary action.

10. FLOW AUTOMATION
Salesforce Flows were implemented to automate important recruitment processes and reduce manual work.
FLOWS IMPLEMENTED

1. Record-Triggered Flow – High Priority Candidate Application Notification
This flow automatically triggers when a candidate application is created or updated.
If the AI Candidate Score is greater than a defined threshold, an email alert is sent to notify the recruiter about a high priority candidate.

2. Record-Triggered Flow – High Salary Offer Submission for Approval
This flow automatically submits the candidate application for approval when the offer amount exceeds the predefined limit.
Condition:
Offer Amount > 1,000,000
The system then sends the record to the hiring manager for approval.

3. Scheduled-Triggered Flow – Pending Candidate Application Follow-Up
This flow runs on a scheduled basis and checks for candidate applications that remain in the pending state for a long time.
If such records are found, the recruiter receives a reminder email to review the application.

4. Screen Flow – Candidate Application Creation Wizard

A screen flow was created to simplify the candidate application creation process.
This wizard allows recruiters to:

• Enter candidate details
• Select job opening
• Add AI candidate score
• Enter proposed offer amount

The flow then creates a new Candidate Application record automatically.

11. AGENTFORCE AI IMPLEMENTATION

Agentforce AI was integrated to assist recruiters and hiring managers in making intelligent hiring decisions.
The Recruitment Assistant Agent was created to analyze candidate data and provide automated insights.

Agent Capabilities

The AI agent can perform the following tasks:

• Generate candidate application summaries
• Analyze hiring risks based on experience and AI score
• Provide recommendations for salary offer approvals
• Assist recruiters in evaluating candidate suitability

Prompt Templates Created:

1. Candidate Application Summary Prompt

Generates a summary of the candidate application including job role, experience level, AI score, and application status.

2. Hiring Risk Analysis Prompt

Analyzes the candidate profile and determines the hiring risk level based on:

- Experience level
- AI candidate score
- Proposed salary
- Job role requirements

3. Offer Approval Guidance Prompt

Provides recommendations to hiring managers regarding candidate offer approval decisions.
The AI agent evaluates the candidate application and suggests whether the offer should be approved or rejected.

12. TESTING AND RESULTS

Test Case 1: AI Candidate Score Validation

Objective:
To ensure that the AI Candidate Score does not exceed the maximum allowed value.

Test Input:
AI Candidate Score = 120

Expected Result:
System should display a validation error.

Actual Result:
The system displayed the message:
"AI Candidate score cannot exceed 100."

Test Case 2: Validation Rule – Offer Amount

Objective:
To ensure that the AI Candidate Score does not exceed the maximum allowed value.

Test Input:
AI Candidate Score = 120

Expected Result:
System should display a validation error.

Actual Result:
The system displayed the message:
"AI Candidate score cannot exceed 100."

Test Case 3: Agentforce AI Summary Generation

Objective:
To verify that the AI agent generates a candidate summary.

Test Input:
Candidate Application Record: **CA-0004**

Expected Result:
The AI agent should generate a summary of the candidate application.

Actual Result:
The system generated a candidate summary including:

- Candidate Name
- Job Opening
- AI Candidate Score
- Offer Amount
- Application Status

Test Case 4: High Priority Candidate Notification

Objective:
To verify that the system sends a notification when a candidate has a high AI score.

Test Input:
AI Candidate Score = 90

Expected Result:
Recruiter should receive an email notification.

Actual Result:
The email notification was successfully triggered.

13.BENEFITS OF THE SYSTEM

The system provides several benefits:
• Faster recruitment decisions
• AI-based candidate evaluation
• Reduced manual analysis
• Automated approval workflows
• Improved transparency in hiring decisions

14. CONCLUSION

The AI-Powered Job Recruitment & Candidate Management System was successfully implemented using Salesforce Admin tools and Agentforce AI.

The system automates key recruitment processes such as candidate application management, salary approval workflows, notifications, and AI-based hiring analysis.

By integrating validation rules, flows, approval processes, email alerts, and AI agents, the system improves recruitment efficiency and reduces manual effort.

The AI agent assists recruiters and hiring managers by generating candidate summaries, analyzing hiring risks, and providing offer approval guidance.

Overall, the system provides a centralized and intelligent platform for managing the recruitment lifecycle and supports better hiring decisions.

15. FUTURE ENHANCEMENTS

Although the system currently supports core recruitment automation, several improvements can be implemented in the future.
• Integration with external job portals such as LinkedIn or job boards
• Resume parsing using AI for automatic candidate data extraction
• Advanced AI models for candidate ranking and predictive hiring success
• Dashboard and analytics for recruitment performance tracking
• Mobile accessibility for recruiters and hiring managers
• Integration with HR management systems for employee onboarding

These enhancements will further improve the system’s intelligence, scalability, and overall recruitment efficiency.

16. BUSINESS CORRESPONDENCE

Effective communication between recruiters, hiring managers, and candidates is essential in the recruitment process.

The system supports business correspondence through automated email notifications and alerts. Email templates and alerts ensure that all stakeholders receive timely updates regarding candidate applications, approval requests, and follow-ups.

Key communications supported in the system include:

- Notification for high priority candidate applications
- Manager approval request for high salary offers
- Follow-up reminders for pending candidate applications
- Communication between recruiters and hiring managers during the hiring process

This automated communication helps maintain transparency, improves coordination among stakeholders, and ensures timely decision-making.

17. TEAM MEMBERS AND RESPONSIBILITIES

SYED HAMISH NEHAL

Responsibilities:

- Custom objects creation
- Fields and relationships configuration
- Approval process implementation
- Flow automation configuration
- Agentforce AI implementation
- Email template creation
- AI agent configuration and testing
- Lightning App configuration
- Profile, Role and Role Hierarchy setup
- Testing
- Documentation preparation

VAISHNAVI

Responsibilities:

- Page layout configuration
- UI customization for objects and record pages

SAURABH KUMAR (TEAM LEAD)

Responsibilities:

- Creation of validation rules
- Configuration of tabs
- Contribution to approval process setup
- Email alerts configuration
- Duplicate and Matching rules implementation

UJJWAL

Responsibilities:

- Configuration and implementation of Dynamic Forms

PRIYANSHU

Responsibilities:

- User creation and user management